

Robotic CyberKnife Radiosurgery for the Treatment of Choroidal and Ciliary Body Melanoma



RAFFAEL LIEGL, VALERIE SCHMELTER, CHRISTOPH FUERWEGER, FELIX EHRET, SIEGFRIED PRIGLINGER, ALEXANDER MUACEVIC, AND PAUL FOERSTER

• **PURPOSE:** To present the results of robotic-assisted radiosurgery in choroidal and ciliary body melanomas treated at the Department of Ophthalmology of Ludwig-Maximilians-University and the European CyberKnife Center in Munich, Germany.

• **DESIGN:** Interventional case series

• **METHODS:** This retrospective study included 594 consecutive patients referred to our clinic for the treatment of choroidal and ciliary body melanomas with robotic radiosurgery (CyberKnife) from 2005 to 2019. Eye retention, local control and disease-specific survival rates were calculated as Kaplan–Meier and actuarial estimates. The impact of prescription dose, tumor size, and ciliary body involvement was assessed by likelihood ratio tests and Cox regression.

• **RESULTS:** Among all patients who were staged according to the TNM classification system (8th edition), 22.7% were I, 57.9% were II, 18.9% were III, and 0.5% were IV. Median apical tumor height and base diameter were 5.8 and 11.4 mm. The mean follow-up was 41.7 months. Local control after 3 and 5 years was 92.0% (95% CI = 88.2%-94.7%) and 84.3% (95% CI = 77.9%-89.0%), respectively, for 21 to 22 Gy and 86.9% (95% CI = 79.7%-91.7%) and 77.7% (95% CI = 68.5%-84.6%), respectively, when treated with 20 Gy or less. Eye retention was achieved in 89.9% and 81.0% after 3 and 5 years with 21 to 22 Gy and 85.9% and 80.0% for 20 Gy or less. Disease-specific survival rates were 93.1% (95% CI = 90.2%-95.2%) after 3 years, 89.8% (95% CI = 86.0%-92.6%) after 5 years, and 87.8% (95% CI = 82.8%-91.4%) after 7 years.

• **CONCLUSIONS:** This is the largest series of patients treated for choroidal and ciliary body melanomas with CyberKnife. Our results reflect an improvement in the outcome of CyberKnife therapy for patients with choroidal

and ciliary body melanoma treated with single-session radiosurgery in the last decade. (Am J Ophthalmol 2023;250: 177–185. © 2022 Published by Elsevier Inc.)

UVEAL MELANOMA IS ALMOST EXCLUSIVELY TREATED with some form of irradiation if enucleation can be avoided. Brachytherapy has emerged to become the most widely used treatment option in choroidal melanoma, and the Collaborative Ocular Melanoma Study Group (COMS) has extensively contributed to our understanding and knowledge in the management of uveal melanoma with plaque radiotherapy.^{1,2} However, restrictions such as tumor size and tumor location exist and limit the use of brachytherapy in uveal melanoma. Ruthenium-106 (Ru-106) plaques, for example, are usually considered in melanomas with an apical height not exceeding 7 mm. Tumors located directly beneath the fovea or in close proximity to the optic nerve are also not very suitable for treatment with radioactive plaques.^{3–5} External beam radiation (EBR) approaches have existed for almost as long as brachytherapy but are usually considered only in cases when brachytherapy is not a suitable treatment option. The linear accelerator (LINAC) is 1 such EBR device and is based on the generation of photons through the acceleration of charged particles. It is also the most commonly used EBR option in the treatment of patients with any form of cancer and is thus available in most major hospitals throughout the world. Treatment of uveal melanoma using a LINAC is a proven treatment option, and several reports have documented similar outcomes regarding local control and overall survival in comparison with the standard of care, proton beam therapy (PBT), in uveal melanoma.^{6–9} In addition, the spectrum and frequency of complications following radiation therapy using photons is also similar to PBT.^{10–13}

When treatment with a LINAC is carried out with a single fraction, it is usually referred to as stereotactic radiosurgery (SRS), a widely used and available method for the treatment of tumors in all parts of the body including brain tumors and metastases.^{14,15} Stereotactic radiotherapy (SRT) uses the same equipment and technical devices but is performed over several days and usually uses smaller radiation fractions per session. It is unclear, however, whether a single radiation session with a higher radiation dose as in

Raffael Liegl is now affiliated with the Department of Ophthalmology at University of Bonn, Bonn, Germany. Felix Ehret is now affiliated with the Department of Neuroradiology at Charité Berlin, Berlin, Germany. Accepted for publication December 16, 2022.

From the Department of Ophthalmology (R.L., V.S., S.P., P.F.), Ludwig-Maximilians University Munich, Munich, Germany; European Radiosurgery Center Munich (C.F., F.E., A.M.), Munich, Germany; Center for Neurosurgery (C.F.), Department of Stereotaxy and Functional Neurosurgery, University of Cologne, Cologne, Germany

Inquiries to Raffael Liegl, University of Bonn, Bonn, Germany; e-mail: raffael.liegl@ukbonn.de

SRS offers the same treatment results as hypofractionated radiation treatment over several days as in SRT.

Frameless robotic radiosurgery offers some distinct advantages over other stereotactic procedures, as it does not require an invasive head frame and can achieve highly conformal treatment plans also for non-intracranial targets such as the eye because of the robotic flexibility. Robotic radiosurgery using the Cyberknife system is based on the aforementioned technique using a LINAC as source of photons for tumor treatment. This LINAC is mounted on a robotic arm to navigate around the treatment target, allowing radiation treatment from virtually all positions. Because radiation treatment delivery can be facilitated from almost every angle, the radiation dose could theoretically be lower and potentially result in a favorable complication rate. However, only a few reports have been published so far on the outcomes of patients with choroidal melanoma treated with robotic radiosurgery.^{10,16}

In this report, we share our experience in the treatment of 594 patients diagnosed with uveal melanoma using stereotactic radiosurgery (SRS) with robotic radiosurgery with respect to local control, tumor size, eye retention, complications, and disease-related death.

METHODS

Consecutive patients diagnosed with uveal melanoma and treated with CyberKnife robotic radiosurgery between 2005 and 2019 were reviewed with approval of the Institutional Review Board ("Ethikkommission der LMU"). The study adhered to the tenets of the Declaration of Helsinki.

A minimum of 1 follow-up examination or a reported adverse event before the first scheduled follow-up was required to be included in this study.

We included all tumors that fulfilled our criteria for the diagnosis of a choroidal melanoma related to known risk factors as described by Shields et al.^{17,18} The diagnosis of a uveal melanoma was based on these factors with emphasis on tumor height and internal reflectivity measured using standardized A-scan echography as well as the appearance of orange pigment. In the vast majority of cases, no doubts regarding the diagnosis of uveal melanoma were present. In cases in which a diagnosis could not be safely established, often because of lower tumor height, patients were either offered a biopsy or were followed up within 3 months. If tumor height increased during this time or other risk factors were additionally found, the case was then discussed among tumor experts, and either a consent diagnosis of uveal melanoma was established or further follow-up appointments were scheduled.

We excluded tumors that had been previously treated by a different method. We did not exclude patients for reasons such as a certain tumor location within the eye, reduced general condition, or status of the fellow eye.

Treatment parameters and clinical data were analyzed, with eye preservation, local tumor control, disease-specific survival, and radiation-related ocular complications as main outcome measures.

Upon first presentation, all patients received a full ophthalmologic examination, including assessment of best-corrected visual acuity, slitlamp biomicroscopy, funduscopy, standardized A-/B-scan ultrasound and/or ultrasound biomicroscopy (UBM), and photographic documentation. All patients received a complete tumor staging by a medical oncologist including ultrasound examination of the liver, assessment of liver enzymes in the blood, and abdominal/thoracic CT scan if indicated at baseline. All patients were thoroughly informed about the chances and risks of the different treatment options available, and informed written consent was obtained before treatment. Treatment options discussed were plaque radiotherapy when applicable (eg, apical tumor height 7.0 mm or less), proton beam therapy, or enucleation. If patients diagnosed with uveal melanoma chose to be treated with PBT, we referred these patients to a different treatment center, and hence these patients were not routinely followed up by us, as we do not offer PBT in our own facility. Patients who were treated with brachytherapy were not included in this series.

CyberKnife radiosurgery was performed as a standardized outpatient procedure as described previously, with some adaptation since our initial report.¹⁹ First, retrobulbar anesthesia was performed to achieve akinesia of the globe within the orbit. Axial T2 and gadolinium-contrast-enhanced T1 magnetic resonance imaging (MRI) of the orbit was performed, followed by a planning computed tomography (CT) scan of the patient head with 1.0-mm axial slices. CT and MRI scans were overlaid for target volume definition by both ophthalmologists and radiation oncologists. Clinical data such as ultrasound or fundus images were also incorporated into treatment planning. A 1-mm margin in all directions and a 2-mm margin posteriorly was added to the visible tumor to create a planning target volume (PTV) intended to compensate for a potential slight posterior shift of the eyeball during treatment due to resorption of the anesthetic. The contralateral eye was blocked for beam traversal. A non-isocentric inverse algorithm was used in cooperation with a medical physicist to identify optimal beam directions and weights (Multiplan and Precision treatment planning software, Accuray Inc). Radiation was delivered in a single fraction with a CyberKnife system (G4: 2005-2012; M6: 2013-2018; Accuray Inc), a dedicated radiosurgery platform for non-isocentric, non-coplanar beam delivery. The system comprises a 6-MV linear accelerator (LINAC) mounted to a 6-axis robotic manipulator and a stereoscopic X-ray imaging system. For treatment, the patient was positioned on a robotic treatment couch, using a thermoplastic mask to stabilize the head. The position of the skull was monitored in stereoscopic X-ray images periodically throughout the treatment. The most recent position information was used to actively guide the treat-

ment beam to the current tumor location. Radiation delivery was completed within 3 hours from retrobulbar anesthesia to ensure immobility of the eye during the whole treatment.

The prescription dose was uniformly set to 21 Gy at the 70% isodose in CyberKnife treatments after 2012. However, patients who were treated before 2012 received prescription doses that were often lower than that. We therefore divided our results into these 2 groups.

Ophthalmologic follow-up examinations were performed every 3 months within the first 12 months and every 6 months thereafter with special focus on tumor control, defined as either continuous regression or no further progression of the melanoma, and radiation-related complications. Oncologic re-staging to assess metastasis was performed twice a year.

Tumor recurrence was defined as any degree of documented tumor growth (in thickness or base) appreciated by ophthalmoscopy and standardized A-scan ultrasonography. Melanoma-related deaths and deaths from other causes were recorded. We further divided all patients according to their American Joint Committee on Cancer (AJCC) staging category into 4 stages (I-IV) and analyzed whether there were differences regarding their survival rates. For analyses, patients in stage IV were analyzed together with those in stage III because of the low number of patients in stage IV (n = 3).

We distinguished between patients who were initially treated with radiation doses of 20 Gy or less and patients who received 21 to 22 Gy in a subgroup analysis.

Only little information regarding outcomes of robotic radiosurgery is available. Multiple reports have shown that treatment outcome is dependent on tumor height and size. To take this aspect into account, we divided our patient cohort into 2 subgroups. Patients who were classified as T1 or T2 according to the AJCC T classification were considered small and medium-sized tumors, as opposed to patients with tumors classified as T3 or T4 who were assigned to the group of large and very large tumors.

Data were collected and analyzed in Excel (Microsoft Corporation). Statistical analysis was performed in STATA/IC 10.1 (StataCorp).

Eye retention, local control, and disease-specific survival rates were calculated by Kaplan–Meier analysis and actuarial estimates. The impact of prescription dose, tumor size, and ciliary body involvement was assessed by likelihood ratio tests and Cox proportional hazards regression analysis. Cox proportional hazards assumption was checked with global goodness-of-fit test based on Schoenfeld residuals.

RESULTS

All patients were categorized following the revised AJCC *Cancer Staging Manual*, 8th edition (Table 1). Median pre-

TABLE 1. Characteristics of Patients Treated With CyberKnife Radiosurgery (N = 594)

Patient sex, male/female, %	51/49
Treatment site, right/left, %	51/49
Patient age, y, median (range)	64.5 (20.9–94.5)
Tumor height, mm, median (range)	5.8 (0.7–16.0)
Largest basal diameter, mm, median (range)	11.4 (1.2–22.0)
COMS, %	
1	9.1
2	70.0
3	20.9
AJCC stage group, %	
I	22.7
II	57.9
III	18.9
IV	0.5
Metastatic development overall, %	14.8
Follow-up, mo, median (range)	35.1 (0.4–156.7)

AJCC = American Joint Committee on Cancer; COMS = Collaborative Ocular Melanoma Study Group.

scription dose to the tumor margin was 21 Gy (range 17–22) enclosing the planning target volume (PTV).

• **PATIENT POPULATION:** A total of 612 patients met the inclusion criteria. Of these, 18 were lost to follow-up. Therefore a total of 594 eyes of 594 patients (305 male, 289 female) with a median age of 64.5 years (mean 63.5, SD 14.0, range 20.9–94.5) were included in this study. The tumor affected the right eye in 51% of patients. The median and mean largest basal tumor diameter was 11.4 and 11.6 mm (range 1.2–22.0) and the median and mean maximal tumor height was 5.8 and 6.2 mm (range 0.7–16.0 mm), respectively. The reflectivity had a median and mean of 50% and 50.0% (range 0–95%). The median and mean follow-up at time of analysis was 35.1 and 41.7 months (range 0.4–156.7 months), respectively. A total of 513 patients had a follow-up of at least 1 year, 288 of 3 years, and 149 of 5 years or longer (Table).

• **TREATMENT RESULTS:**

Local control

Local tumor control after single-session radiosurgery was achieved in 514 of 594 eyes (86.5%) within the observation period. Recurring tumors were treated by enucleation (68%) or repeat radiosurgery (28%). One patient decided to seek treatment elsewhere after being diagnosed with a recurrence of his melanoma, and 1 patient was scheduled for another CyberKnife treatment but was not treated by the time that the final analysis was performed. Kaplan–Meier estimated rates of local control at 3, 5, and 7 years were 90.7% (95% CI = 87.5%–93.1%), 82.4% (95% CI = 77.4%–86.4%), and 69.0% (95% CI = 60.9%–75.8%), respectively. When comparing patients who received 21 to 22 Gy at a 70% isodose to pa-

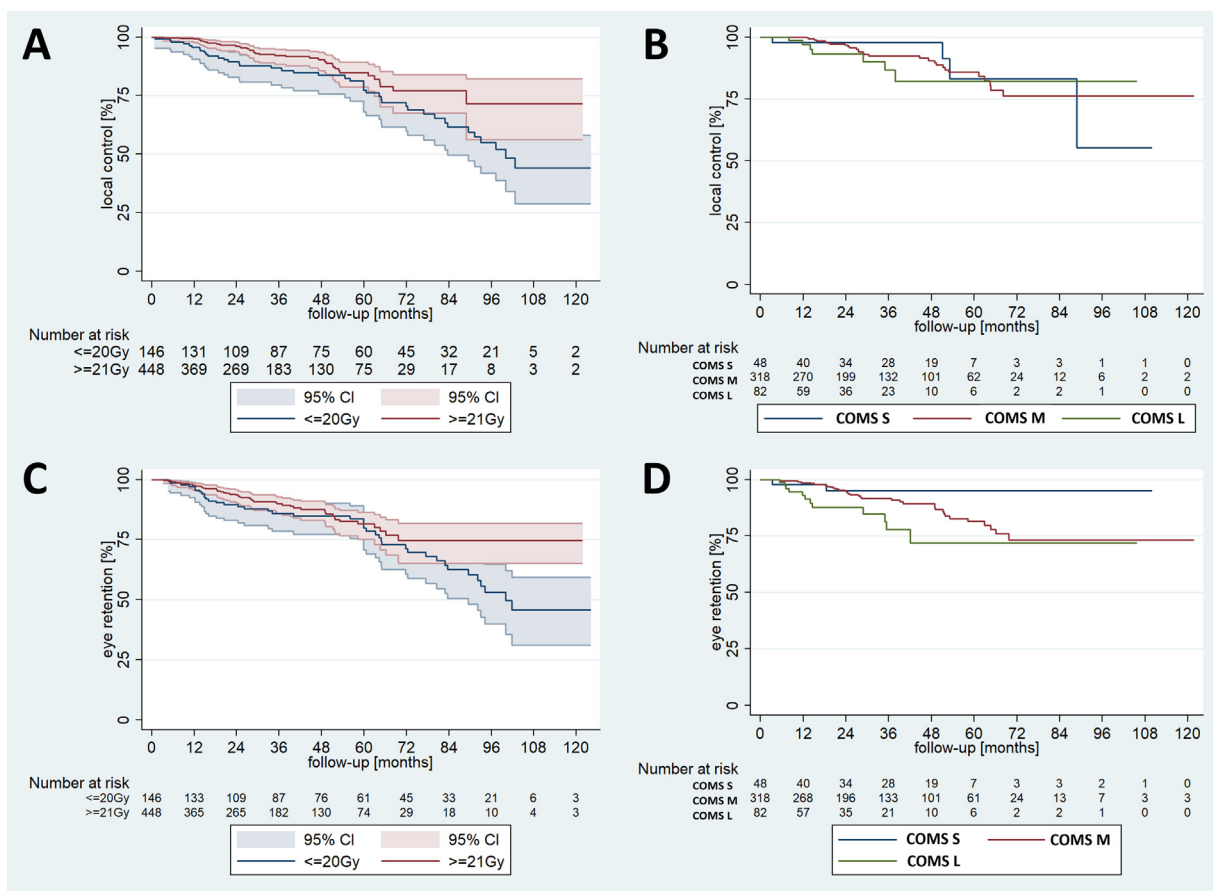


FIGURE 1. A. Kaplan–Meier estimated local control in patients treated with either at least 21 Gy or 20 Gy and less. B. Kaplan–Meier estimated local control in patients treated with at least 21 Gy grouped by tumor size according to the Collaborative Ocular Melanoma Study Group (COMS). C. Kaplan–Meier estimated eye retention in patients treated with either at least 21 Gy or 20 Gy and less. D. Kaplan–Meier estimated eye retention in patients treated with either at least 21 Gy grouped by tumor size according to COMS.

tients with a prescription dose of 20 Gy or less, significant differences were found. Local control for 21 to 22 Gy was achieved in 409 of 448 patients overall (91.3%) and was 92.0% (95% CI = 88.2%-94.7%) after 3 years, 84.3% (CI = 77.9%-89.0%) after 5 years, and 76.6% (95% CI = 67.2%-83.6%) after 7 years. A total of 146 patients received 20 Gy or less, and in this group, 41 patients developed local recurrences. This resulted in an overall local control of 71.9%; local control was achieved in 86.9% (95% CI = 79.7%-91.7%) after 3 years, 77.7% (CI 68.5%-84.6%) after 5 years, and 62.1% (95% CI = 50.3%-71.8%) after 7 years (Figure 1, A). A likelihood ratio test confirmed significantly differing results between the 2 groups ($P = .0008$).

Size dependence of local control rates were assessed by dividing patients receiving 21 to 22 Gy into 3 groups according to COMS as well as 2 groups according to T categories (small and medium-sized T1+T2 vs large and very large T3+T4 tumors).

Classification according to COMS yielded 3-, 5-, and 7-year local control rates of 97.8% (95% CI = 85.1%-

99.7%), 83.8% (95% CI = 54.4%-95.0%), and 83.8% (95% CI = 54.4%-95.0%) for small, 92.1% (95% CI = 87.3%-95.1%), 85.1% (95% CI = 77.8%-90.2%), and 75.8% (95% CI = 64.7%-83.8%) for medium, and 87.1% (95% CI = 72.4%-94.3%), 82.0% (95% CI = 63.1%-91.8%), and 82.0% (95% CI = 63.1%-91.8%) for large tumors (Figure 1, B). Results did not differ significantly between the groups ($P = 0.72$, likelihood ratio test).

Local control rates for small and medium-sized T1+2 tumors were 93.8% (95% CI = 89.6%-96.4%), 86.8% (95% CI = 79.5%-91.6%), and 77.1% (95% CI = 65.8%-85.1%) after 3, 5, and 7 years. Large and very large tumors (T3+4) had local control rates of 87.2% (95% CI = 77.0%-93.1%), 77.0% (60.4%-87.3%), and 77.0% (95% CI = 60.4%-87.3%) after 3, 5, and 7 years, respectively. The results did not differ significantly between the 2 groups ($P = .42$, likelihood ratio test).

Cox regression confirmed that the prescription dose had had a significant impact on local control (HR = 0.73, 95% CI = 0.56-0.95, $P = .018$), whereas tumor size (T1-T4;

HR = 1.16, 95% CI = 0.87-1.56, $P = 0.31$) and involvement of the ciliary body (HR = 1.27, 95% CI = 0.76-2.12, $P = .36$) had not.

Eye retention

Overall, eye retention was accomplished in 506 patients (85.2%) within the observation period (Figure 1, C). The main causes of enucleation were tumor recurrence (59%), secondary glaucoma (38%), and other complications (3%). Actuarial eye retention rates were 88.8% (95% CI = 85.4%-91.4%) at 3 years, 81.0% (95% CI = 76.1%-85.1%) after 5 years, and 68.0% (95% CI = 60.1%-74.6%) after 7 years. When stratified for the prescribed dose, eye retention rates were varying accordingly ($P = .018$, likelihood ratio test). The eye retention rates for patients receiving 20 Gy or less were 85.9% (95% CI = 78.5%-90.9%), 80.0% (95% CI = 71.2%-86.4%), and 62.8% (95% CI = 51.0%-72.6%) after 3, 5, and 7 years. In contrast, patients with higher doses (≥ 21 Gy) had a higher overall eye retention, with 89.9% (95% CI = 85.9%-92.8%), 81.0% (95% CI = 74.5%-86.1%), and 75.0% (95% CI = 66.2%-81.8%) after 3, 5, and 7 years, respectively.

Size dependence of eye retention was assessed by dividing patients receiving 21 to 22 Gy into 3 groups according to COMS, as well as 2 groups according to T categories (small and medium-sized T1+T2 vs large and very large T3+T4 tumors).

Classification according to COMS yielded 3-, 5-, and 7-year eye retention rates of 95.2% (95% CI = 81.9%-98.8%), 95.2% (95% CI = 81.9%-98.8%), and 95.2% (95% CI = 81.9%-98.8%) for small, 91.5% (95% CI = 86.9%-94.5%), 80.8% (95% CI = 72.8%-86.6%), and 73.5% (95% CI = 63.0%-81.5%) for medium, and 78.3% (95% CI = 63.0%-87.9%), 73.4% (95% CI = 55.4%-85.1%), and 73.4% (95% CI = 55.4%-85.1%) for large tumors. COMS groups differed significantly ($P = .023$, likelihood ratio test) (Figure 1, D).

Analogously, tumors of categories T3+4 caused a significantly lower chance of eye retention compared to T1+2 ($P = .034$). T1+2 tumors resulted in eye retention rates of 92.8% (95% CI = 88.5%-95.6%), 84.7% (95% CI = 77.3%-89.9%), and 77.0% (95% CI = 66.3%-84.8%) after 3, 5, and 7 years. For T3+4 tumors, the rates were 83.1% (95% CI = 73.1%-89.6%), 71.2% (95% CI = 55.7%-82.1%), and 71.2% (95% CI = 55.7%-82.1%).

In contrast to the univariate results, Cox regression showed that only tumor size had a significant impact on eye retention (T1-T4, HR = 1.52, 95% CI = 1.16-2.01, $P = .003$), whereas an impact of prescription dose (Gy, HR = 0.86, 95% CI = 0.66-1.12, $P = .28$) and involvement of the ciliary body (HR = 0.96, 95% CI = 0.59-1.59, $P = .89$) could not be confirmed.

- **SURVIVAL:** Of 594 patients, 82 (13.8%) died. A total of 40 (48.8%) died due to progression of their choroidal or ciliary body melanoma, 36 (43.9%) due to non-disease re-

lated causes, and 6 (7.3%) for unknown reasons. Disease-specific survival rates were 93.1% (95% CI = 90.2%-95.2%) after 3 years, 89.8% (95% CI = 86.0%-92.6%) after 5 years, and 87.8% (95% CI = 82.8%-91.4%) after 7 years (Figure 2). Classification according to AJCC yielded 3-, 5-, and 7-year disease-specific survival rates of 100%, 98.7% (95% CI = 91.1%-99.8%) and 94.6% (95% CI = 76.6%-98.9%) for stage I, 91.5% (95% CI = 87.1%-94.4%), 88.0% (95% CI = 82.6%-91.9%), and 88.0% (95% CI = 82.6%-91.9%) for stage II, and 89.3% (95% CI = 78.6%-94.8%), 81.8% (95% CI = 65.8%-90.9%), and 75.8% (95% CI = 55.2%-87.9%) for stage III+IV, respectively (Figure 2). Stages I to III+IV differed significantly ($P = .0001$, likelihood ratio test). Because of the low number of patients in stage IV ($n = 3$), we decided to include these patients in the group of stage III patients.

- **COMPLICATIONS:** The spectrum of radiation-related side effects was similar to complications reported from other radiation treatment options. Serous retinal detachment was observed in 216 patients (36.4%) after treatment. In 167 patients (28.1%), serous detachment was present at the time of initial diagnosis. Another 39 patients developed serous retinal detachment following CyberKnife therapy. Secondary glaucoma occurred in 104 patients (17.5%), of whom 40 (38.5%) had to be enucleated because of intractable secondary glaucoma. A total of 44 patients (42.3%) needed other forms of surgical management, and 19 (18.3%) were sufficiently treated with medication alone. Vitreous hemorrhage occurred in 74 patients (12.5%). Radiation retinopathy occurred in 72 patients (12.1%) during the observation time.

DISCUSSION

We report on 594 patients diagnosed with uveal melanoma and treated with image-guided robotic radiosurgery (CyberKnife, Accuray Inc). The median apical height and largest basal diameter in these patients at initial diagnosis were 5.8 mm and 11.4 mm, respectively. The 3-year local control rates were 92.0% in patients receiving at least 21 Gy to the 70% isodose and 86.9% in the group of patients treated with less than 21 Gy. The 5-year local control rates were 84.3% in patients treated with at least 21 Gy and 77.7% in patients receiving less than 21 Gy (Figure 1A). The distinction between these 2 groups has evolved over the 15-year experience with the CyberKnife system. It is a typical feature of all newer treatment modalities that changes in the treatment algorithm are based upon experience. In recent years, we have identified reasons that have led to inadequate treatment outcome. Among these, subliminal radiation dose to the tumor was the most obvious cause, so that today treatment includes a prescribed radiation dose of 21 Gy at an isodose of 70%.

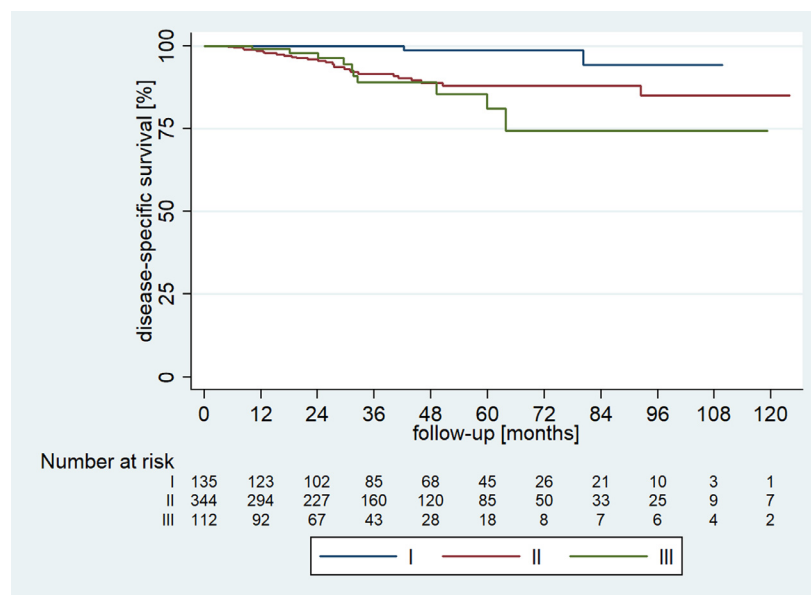


FIGURE 2. Kaplan–Meier estimated disease-specific survival for stages I, II, and III following the *AJCC Cancer Staging Manual*, 8th edition. Because of the low number of patients diagnosed with stage IV ($n = 3$), these patients were included in the group of stage III patients for statistical analyses.

Many studies have reported clinical outcomes, including local control rate as an important marker of therapeutic efficacy. A recent review collected studies investigating the efficacy of Ru-106 in uveal melanoma with cohorts of more than 30 patients. It was found that local control rates ranged from 59% to 98% in patients with a mean apical tumor height of 3.02 mm to 9.29 mm.²⁰ Patients in our cohort had a median apical height of 5.8 mm. When comparing reports from patients with tumors of similar median apical height, local control was reported to be 86.5% for tumors with a height of 4.8 mm²¹ and 84.0% for tumors with an apical height of 5.33 mm.²² The local control rate was 82.4% for particularly large tumors with a thickness of 8.12 mm.²³ For I-125 plaques, the data are even more reliable, as they can be retrieved from COMS Report No. 19. In this report, treatment failure occurred in 10.3% of cases with medium-sized tumors within the first 5 years.¹ Although larger tumors were not studied in the COMS report, other investigators have done so and have reported high local control rates in these tumors as well.²⁴ Our results are comparable to these studies regarding the local control rate; however, they do not match the best results that have been reported on treatment modalities such as brachytherapy.

For larger tumors or tumors that are located adjacent to the optic nerve or underneath the fovea, external beam radiation (EBR) options are often considered. Among these, no prospective randomized trial has been undertaken comparing survival or treatment efficacy. The most common therapeutic options when talking about EBR are proton beam therapy (PBT) or some form of stereotactic radiotherapy (SRT) often using either LINAC or Gamma

Knife–based approaches.^{25–27} Many retrospective reports have been published on local control and eye retention of uveal melanoma treated with PBT and SRT. For instance, Dunavoelgyi et al treated 212 patients with a linear accelerator delivering photons with 50, 60, or 70 Gy in 5 fractions to tumors with a median height of 4.2 mm. They reported a local control in 95.9% of patients after 5 years.²⁸ Modorati et al evaluated Gamma Knife radiosurgery for uveal melanoma, and found local control in 71 of 78 patients (91%). However, these numbers are difficult to compare to our study results, as the median follow-up time was 31.3 months in this study and no 3- or 5-year local control rates were presented.²⁹ When irradiation is carried out using 1 single fraction, it is referred to as SRS. For SRS, local control has been reported to be between 70% and 95%. Results have to be interpreted with caution, as local control rates have often been presented with inconsistent long-term follow-up. Thus, it is difficult to compare the actual rates.^{8–10,12,16} In addition, patient cohorts have often not been uniform among the different studies, and radiation delivery has been very variable. Typically, treatment results are always dependent on the tumor size, because larger tumors carry a worse prognosis. This also applies to reports on treatment outcomes after PBT. Local tumor control rates vary between 67%, as reported by Conway et al for particularly large tumors treated, with a median size of more than 8 mm, to above 95% by several other groups who described the treatment outcomes of tumors of smaller sizes.^{27,30,31} No prospective randomized direct comparison of SRS and PBT has been performed, and the only retrospective non-randomized study was published by Sikuade et al. They

compared SRS to PBT and found PBT to be superior in regard to preservation of visual acuity. SRS performed slightly better regarding local control. Nevertheless, the authors addressed several limitations regarding the comparability between the 2 groups.³²

Our data presented in this report regarding local control demonstrated an improvement of local control rates using SRS with CyberKnife, as the treatment modality has been adjusted on the grounds of our experience. Local control improved from 70% to 84.3% when current settings with the use of 21 Gy are used for the treatment of uveal melanoma. The local control rate of 84.3% that is achieved with the newer setting is similar to the above-mentioned clinical outcomes of other treatment options, such as brachytherapy using Ru-106, as well as other EBR options.

Insufficient local control may result in increased risk of metastasis and subsequent risk of disease-related death.^{33,34} Despite the importance of maximizing local control and the progress with the achievement of local control rates of more than 90%, survival rates for patients with uveal melanoma have not changed significantly over the past 4 decades. This suggests that other factors, such as genetics, must play an important role with respect to survival.^{35,36} Chromosomal abnormalities such as monosomy 3, 8p loss, or 8q gain are commonly found in uveal melanoma patients and are strongly associated with an increased risk of metastasis.³⁷

According to Shields et al, the predicted 5-year disease-specific mortality rate is 31% to 35% for large tumors, 10% for medium tumors, and 1% for small tumors according to the COMS classification.³⁸ Our patient cohort had an overall 5-year disease-specific survival of 89.8% (Figure 2). This finding goes hand in hand with the mortality rate reported by Shields et al when having the mean tumor size of our patient cohort in mind that would be considered medium sized tumors. When looking at the different stages according to the AJCC staging system, 5-year disease-specific survival was 98.7% for stage I tumors, 88.0% for stage II, and 81.9% for stage III, which included 3 patients who were in stage IV according to AJCC criteria. Our results compare well in stages I and II to findings by the landmark work of Kujala et al, who showed a 5-year survival of 96% in stage I, 81% to 86% in stage II (A+B), but a considerably lower rate of 26% to 66% for stage III (A-C).³⁹

The survival rate in our study compares well to that reported by Macdonald et al, who found a disease-specific survival of 87.7% after treatment with PBT.⁴⁰ It is also in agreement with Modorati et al, who reported a rate of 81.9% at 5 years following Gamma Knife radiosurgery.²⁹ Similar numbers have been reported on survival following brachytherapy.⁴¹

The eye retention rate in the current study was 89.9% for patients treated with at least 21 Gy and 85.9% for 20 Gy or less after 3 years and 80% and 81%, respectively, after 5 years (Figure 1D). These results are considerably bet-

ter than our previously reported results¹⁰ and compare well to results of hypo-fractionated SRT delivery as reported by Yazici et al, in which 73% of patients could keep their eye within the first 2 years after treatment.¹² However, in comparison to the COMS trial, the eye retention rate is still lower, as 12.5% were enucleated after 5 years in this study.¹

Results after PBT show an eye retention rate of 90% after 5 years, but it must be considered that these studies report results after treatment of tumors smaller than those in our study.^{27,30}

When looking at studies with only larger tumors, eye retention decreases to 77.4% after 5 years.⁴² Similar findings were reported by Macdonald et al in 147 patients treated with PBT, of whom 22.4% had to be enucleated after a mean time interval of 2 years after treatment.⁴⁰ Both our results as well as published results on PBT are worse than those in a study that reported an eye retention rate of 84% at 5 years using iodine-125 plaques in large tumors.²⁴ The comparison of all of these studies against each other remains difficult, however, as the group of studied patients is often inhomogeneous.

We pursued a different approach concerning fractionation of EBR. Although fractionation might hypothetically be associated with less secondary complications, single-session radiation therapy in combination with retrobulbar anesthesia for global eye akinesia poses a considerably lower treatment burden. That is why we prefer a single-session radiation approach.

The data regarding complications in our cohort were similar to reported data of other radiation therapies.^{43,44} Secondary glaucoma occurred in 104 patients, of whom 40 had to be enucleated because of intolerable secondary glaucoma. Serous retinal detachment was observed in 216 patients. Of these, 167 were already diagnosed with serous detachment upon first diagnosis, and another 49 patients developed serous detachment after treatment. Vitreous hemorrhage occurred in 74 patients, and radiation retinopathy was documented in 72 patients during the observation time. Higher incidences of radiation-related complications, especially radiation retinopathy, were reported by Yazici et al. They were able to show a proportional association with the median eye dose.¹² Our study cohort had lower incidences, with respect to both secondary glaucoma and radiation retinopathy, in comparison to results recently published by Modorati et al after Gamma Knife radiosurgery.¹³

This study contains the largest patient cohort treated with CyberKnife therapy and as well as thoroughly documented data with a long follow-up interval. Nevertheless, it has some limitations because of its retrospective nature, with partially incomplete data on patient and tumor characteristics and treatment outcomes. The median follow-up time of 43 months is the longest that has been published for SRS and CyberKnife in particular; however, it may still be regarded as rather short for reports on choroidal and ciliary body melanoma treatment outcomes.

Overall, our data show an improvement in terms of local control compared to our first results with single-session robotic-assisted radiosurgery. In this study, we report a local control rate of 84.3% after 5 years and a disease-

specific survival rate of 89.8%. Thus, robotic-assisted CyberKnife therapy represents a good therapeutic option in the treatment of small and large choroidal and ciliary body melanoma.

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